PIONEER 2026 – PRAKALP WINNERS

Fueling Entrepreneurship Through Innovation

KIT's College of Engineering, Kolhapur

Intelligent YOLOv8-Based Rover for Precision Agriculture: Tomato Ripeness and Disease Detection

Event: Prakalp – Pioneer 2026

Cash Prize: ₹20,000

About the Project

Our project integrates a YOLOv8-based computer vision model with an autonomous rover to accurately detect tomato ripeness stages and identify plant diseases in real time. By combining robotics, AI-driven image processing, and precision agriculture principles, the solution aims to reduce manual labor, minimize crop losses, and improve yield quality through early disease detection and optimized harvesting decisions.

Project Members

- Joshin Ranedae J S (6BTEC)
- Logesh A J (6BTEC)
- Hari Narayan L M (6BTEC)
- Tanusharvesh D (4BTEC)

The team was honored with a cash prize of ₹20,000, recognizing the technical innovation and practical impact of the project in advancing smart agricultural solutions.